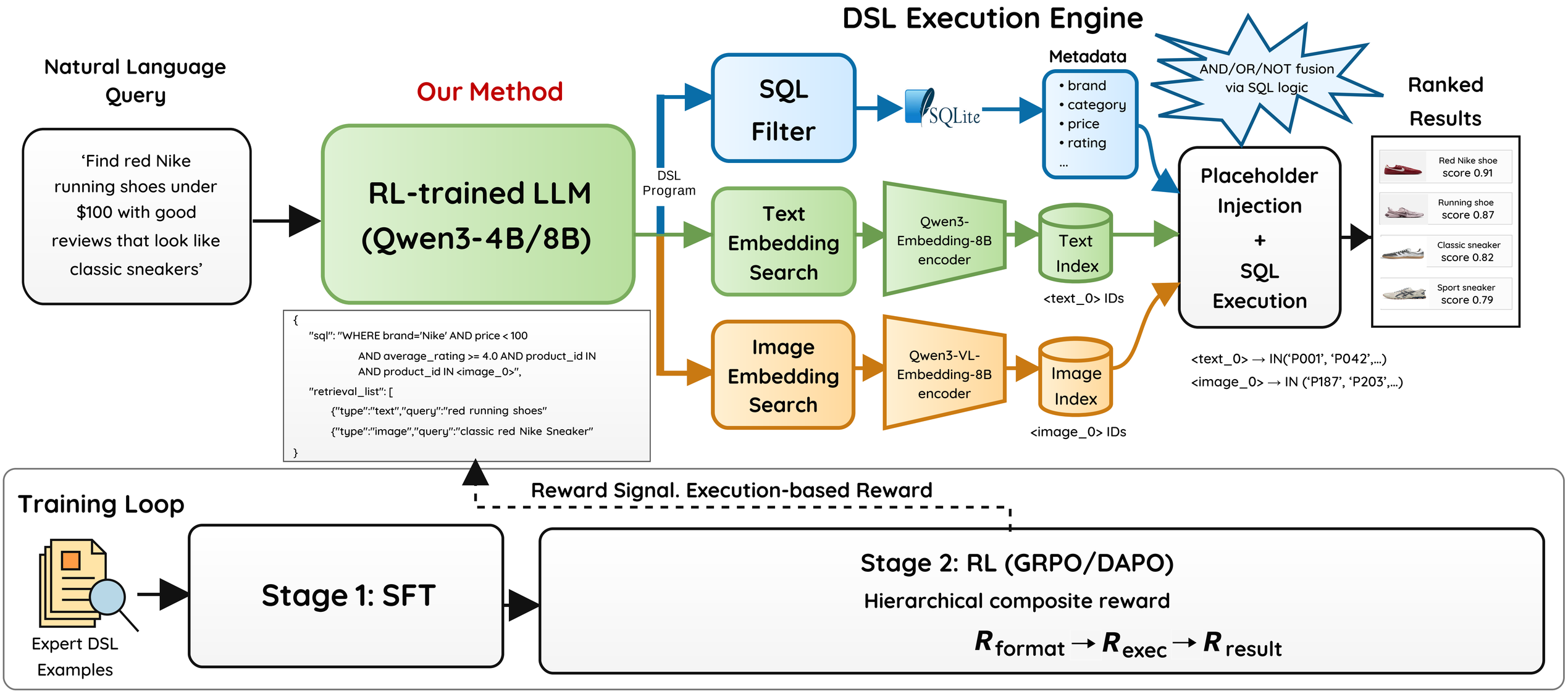

Natural Language Query
'Find red Nike running shoes under $100 with good reviews that look like classic sneakers'
Our Method
RL-trained LLM (Qwen3-4B/8B)
DSL Program
{
"sql": "WHERE brand='Nike' AND price < 100
AND average_rating >= 4.0 AND product_id IN
AND product_id IN <image_0>",
"retrieval_list": [
{"type":"text","query":"red running shoes"
{"type":"image","query":"classic red Nike Sneaker"
}
DSL Execution Engine
SQL Filter
SQLite
Metadata
• brand
• category
• price
• rating
...
AND/OR/NOT fusion via SQL logic
Text Embedding Search
Qwen3-Embedding-8B encoder
Text Index
<text_0> IDs
Image Embedding Search
Qwen3-VL-Embedding-8B encoder
Image Index
<image_0> IDs
Placeholder Injection + SQL Execution
<text_0> → IN('P001', 'P042',...)
<image_0> → IN ('P187', 'P203',...)
Ranked Results
Red Nike shoe score 0.91
Running shoe score 0.87
Classic sneaker score 0.82
Sport sneaker score 0.79
Reward Signal. Execution-based Reward
Training Loop
Expert DSL Examples
Stage 1: SFT
Stage 2: RL (GRPO/DAPO)
Hierarchical composite reward
$R_{format} \rightarrow R_{exec} \rightarrow R_{result}$